Case118 Results:

```
Buses: 118
Lines: 173
Loads: 99
Generators: 53
Baseline max line loading (%): 4.4750873243376645
```

```
Nodes: 118
Edges: 166
```

```
Top 10 nodes by node betweenness
     bus_id  degree  node_betweenness  closeness  eigenvector
76       76       6          0.358312   0.185259     0.235613
48       48       9          0.348221   0.192995     0.295946
68       68       5          0.346251   0.197954     0.263811
79       79       6          0.220349   0.166867     0.210801
69       69       5          0.209553   0.183058     0.164423
23       23       3          0.185385   0.170248     0.051269
22       22       4          0.181974   0.159662     0.016799
99       99       8          0.174114   0.138183     0.301774
41       41       3          0.154862   0.175750     0.091045
39       39       4          0.143979   0.164200     0.038459

Top 10 edges by edge betweenness
     line_id  from_bus  to_bus  edge_betweenness
103      109        68      76          0.272776
77        97        48      68          0.264295
115      114        76      79          0.219882
37        99        23      69          0.183112
34        28        22      23          0.182098
63        62        41      48          0.162837
61        52        39      41          0.146550
57        48        36      39          0.136915
36        37        22      31          0.131903
101       98        68      69          0.130149
```

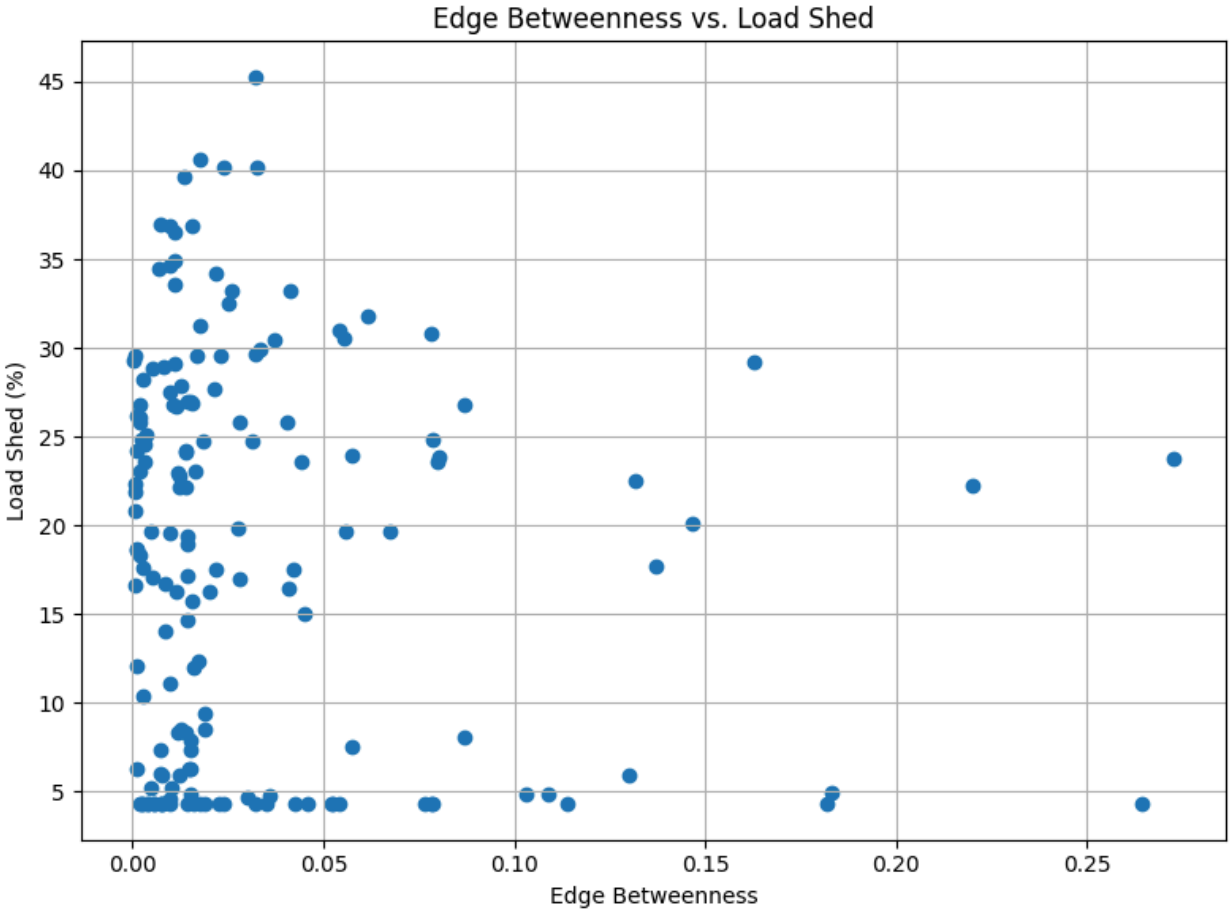
| | initial_line | initial_lines | cascade_size | tripped_lines | iterations | solver_failed | lcc_size | served_load_mw | \ |
|---|---|---|---|---|---|---|---|---|---|
| 19 | 19 | [19] | 99 | [19, 13, 14, 15, 16, 17, 20, 21, 22, 23, 24, 2... | 8 | False | 23 | 2323.0 | |
| 25 | 25 | [25] | 87 | [25, 22, 23, 24, 28, 40, 41, 46, 49, 99, 102, ... | 8 | False | 28 | 2518.0 | |
| 26 | 26 | [26] | 90 | [26, 22, 23, 24, 28, 166, 32, 38, 40, 41, 46, ... | 8 | False | 21 | 2536.0 | |
| 27 | 27 | [27] | 80 | [27, 21, 22, 23, 24, 28, 32, 37, 38, 41, 165, ... | 9 | False | 32 | 2536.0 | |
| 125 | 125 | [125] | 98 | [125, 126, 127, 130, 110, 116, 117, 118, 119, ... | 15 | False | 15 | 2561.0 | |
| 126 | 126 | [126] | 98 | [126, 125, 127, 130, 110, 116, 117, 118, 119, ... | 15 | False | 15 | 2561.0 | |
| 0 | 0 | [0] | 99 | [0, 1, 15, 17, 18, 28, 40, 49, 51, 14, 16, 19,... | 13 | False | 11 | 2675.0 | |
| 58 | 58 | [58] | 87 | [58, 59, 60, 63, 64, 23, 40, 41, 43, 46, 52, 5... | 6 | False | 31 | 2676.0 | |
| 64 | 64 | [64] | 87 | [64, 58, 59, 60, 63, 23, 40, 41, 43, 46, 52, 5... | 6 | False | 31 | 2676.0 | |
| 31 | 31 | [31] | 109 | [31, 35, 36, 38, 39, 168, 23, 28, 29, 40, 49, ... | 13 | False | 11 | 2694.0 | |

| | shed_load_mw | shed_percent |
|---|---|---|
| 19 | 1919.0 | 45.238095 |
| 25 | 1724.0 | 40.641207 |
| 26 | 1706.0 | 40.216879 |
| 27 | 1706.0 | 40.216879 |
| 125 | 1681.0 | 39.627534 |
| 126 | 1681.0 | 39.627534 |
| 0 | 1567.0 | 36.940123 |
| 58 | 1566.0 | 36.916549 |
| 64 | 1566.0 | 36.916549 |
| 31 | 1548.0 | 36.492221 |

| | pearson_betweenness_vs_shed | spearman_betweenness_vs_shed | pearson_betweenness_vs_cascade | spearman_betweenness_vs_cascade |
|---|---|---|---|---|
| 0 | -0.098274 | -0.048692 | -0.107213 | -0.089691 |

Top 15 edge comparison rows

| | line_id | from_bus | to_bus | edge_betweenness | initial_line | cascade_size | shed_percent | lcc_size | solver_failed |
|---|---|---|---|---|---|---|---|---|---|
| 46 | 19 | 14 | 16 | 0.032117 | 19 | 99 | 45.238095 | 23 | False |
| 67 | 25 | 19 | 20 | 0.017775 | 25 | 87 | 40.641207 | 28 | False |
| 55 | 26 | 20 | 21 | 0.024004 | 26 | 90 | 40.216879 | 21 | False |
| 44 | 27 | 21 | 22 | 0.032696 | 27 | 80 | 40.216879 | 32 | False |
| 95 | 126 | 88 | 89 | 0.013690 | 126 | 98 | 39.627534 | 15 | False |
| 125 | 0 | 0 | 1 | 0.007388 | 0 | 99 | 36.940123 | 11 | False |
| 77 | 64 | 47 | 48 | 0.015469 | 64 | 87 | 36.916549 | 31 | False |
| 114 | 58 | 45 | 46 | 0.009759 | 58 | 87 | 36.916549 | 31 | False |
| 105 | 31 | 26 | 27 | 0.011179 | 31 | 109 | 36.492221 | 11 | False |
| 108 | 5 | 5 | 6 | 0.010967 | 5 | 75 | 34.936351 | 32 | False |
| 113 | 50 | 38 | 39 | 0.009766 | 50 | 60 | 34.629892 | 36 | False |
| 128 | 10 | 10 | 11 | 0.006725 | 10 | 72 | 34.464875 | 32 | False |
| 60 | 39 | 26 | 31 | 0.021795 | 39 | 102 | 34.181990 | 13 | False |
| 107 | 115 | 78 | 79 | 0.010982 | 115 | 89 | 33.569071 | 29 | False |
| 53 | 29 | 22 | 24 | 0.026148 | 29 | 77 | 33.262612 | 31 | False |


Edge Betweenness vs. Load Shed
Load Shed (%)
Edge Betweenness
45
40
35
30
25
20
15
10
5
0.00
0.05
0.10
0.15
0.20
0.25

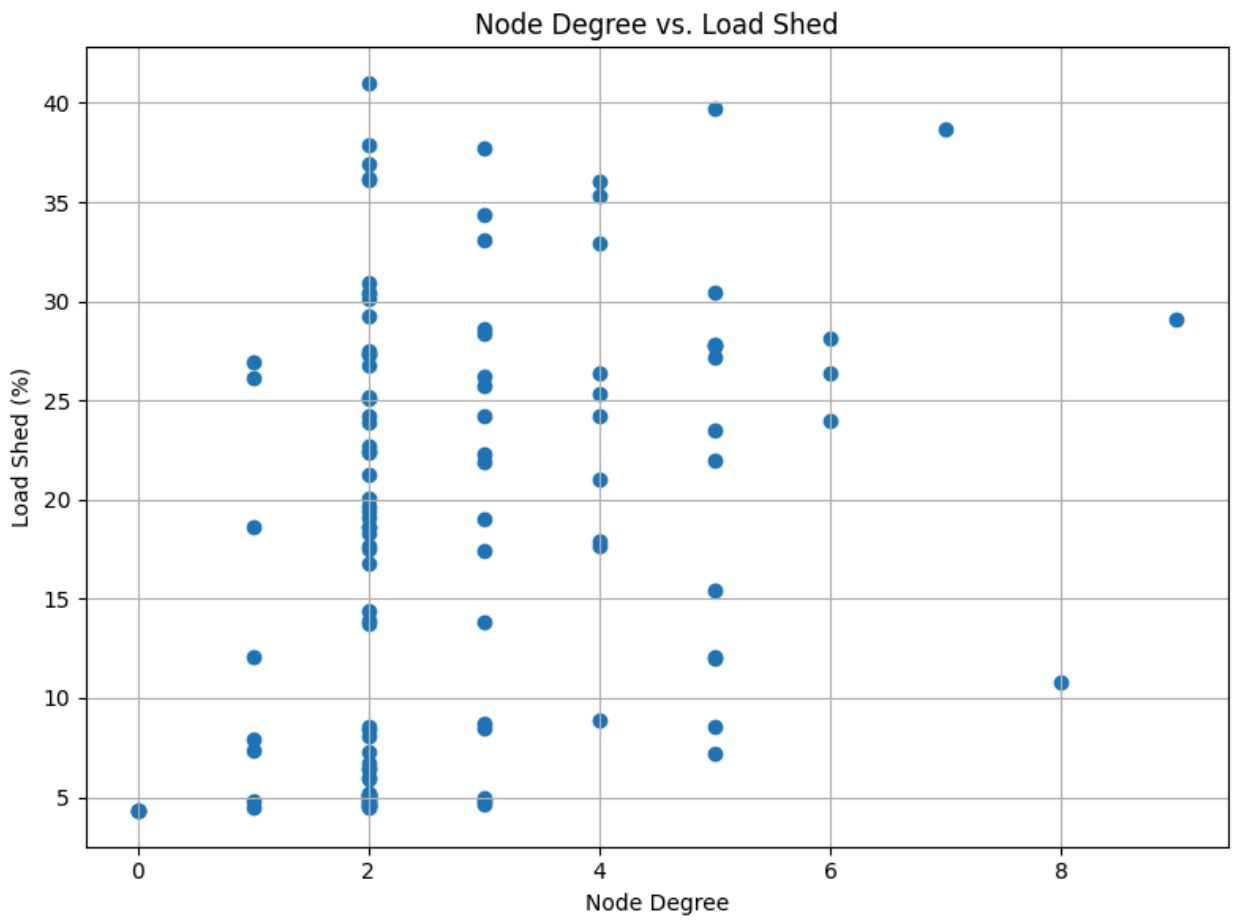
| | initial_line | initial_lines | cascade_size | tripped_lines | iterations | solver_failed | lcc_size | \ |
|---|---|---|---|---|---|---|---|---|
| 20 | NaN | [25, 26] | 91 | [25, 26, 22, 23, 24, 28, 166, 32, 38, 40, 41, ... | 8 | False | 21 | |
| 36 | NaN | [43, 44, 46, 47, 48] | 97 | [43, 44, 46, 47, 48, 13, 14, 15, 16, 17, 19, 2... | 4 | False | 21 | |
| 11 | NaN | [10, 11, 12, 13, 15, 18, 170] | 76 | [10, 11, 12, 13, 15, 18, 170, 0, 1, 3, 17, 20,... | 10 | False | 32 | |
| 89 | NaN | [125, 126, 127] | 97 | [125, 126, 127, 110, 116, 117, 118, 119, 120, ... | 14 | False | 15 | |
| 46 | NaN | [58, 60, 96] | 88 | [58, 60, 96, 59, 63, 64, 23, 40, 41, 43, 46, 5... | 6 | False | 31 | |
| 0 | NaN | [0, 1] | 99 | [0, 1, 15, 17, 18, 28, 40, 49, 51, 14, 16, 19,... | 12 | False | 11 | |
| 7 | NaN | [6, 33] | 93 | [6, 33, 0, 5, 10, 11, 12, 13, 14, 15, 16, 17, ... | 4 | False | 31 | |
| 27 | NaN | [31, 32] | 74 | [31, 32, 36, 38, 39, 168, 23, 28, 35, 167, 169... | 11 | False | 32 | |
| 26 | NaN | [30, 31, 39, 168] | 77 | [30, 31, 39, 168, 23, 25, 28, 29, 34, 35, 36, ... | 5 | False | 33 | |
| 102 | NaN | [150, 152, 153, 161] | 100 | [150, 152, 153, 161, 139, 140, 151, 154, 155, ... | 13 | False | 15 | |
| 44 | NaN | [56, 57, 63] | 95 | [56, 57, 63, 46, 52, 53, 55, 24, 41, 42, 43, 4... | 7 | False | 15 | |
| 60 | NaN | [84, 85, 87] | 70 | [84, 85, 87, 86, 88, 93, 94, 95, 40, 47, 48, 5... | 5 | False | 31 | |
| 10 | NaN | [8, 9, 10, 14] | 69 | [8, 9, 10, 14, 3, 4, 5, 13, 15, 16, 17, 18, 22... | 7 | False | 31 | |
| 1 | NaN | [0, 11] | 68 | [0, 11, 1, 12, 18, 14, 15, 16, 17, 20, 24, 28,... | 8 | False | 31 | |
| 17 | NaN | [21, 22] | 81 | [21, 22, 19, 20, 24, 28, 35, 40, 49, 51, 165, ... | 14 | False | 32 | |

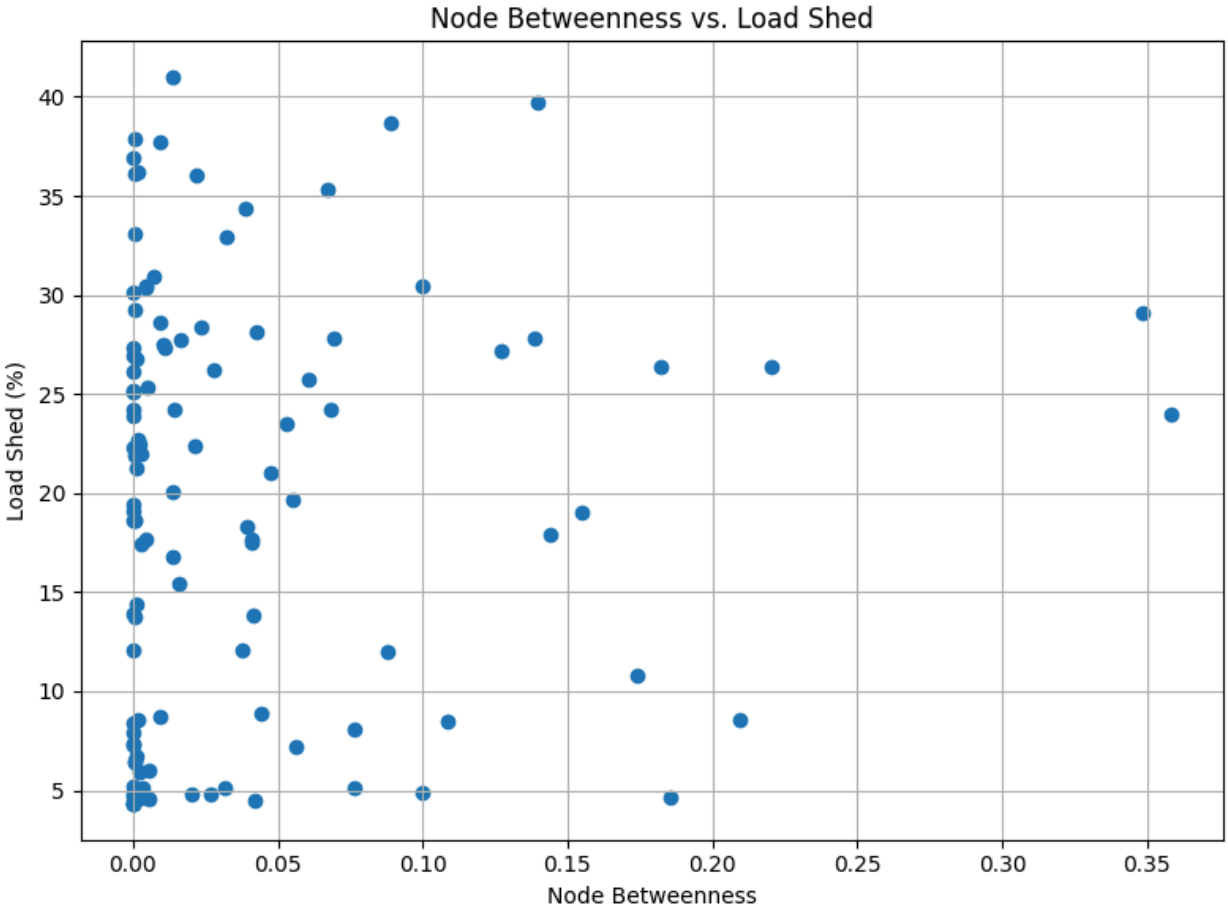
| | served_load_mw | shed_load_mw | shed_percent | initial_bus | incident_line_count |
|---|---|---|---|---|---|
| 20 | 2504.0 | 1738.0 | 40.971240 | 20 | 2 |
| 36 | 2556.0 | 1686.0 | 39.745403 | 36 | 5 |
| 11 | 2601.0 | 1641.0 | 38.684583 | 11 | 7 |
| 89 | 2636.0 | 1606.0 | 37.859500 | 89 | 3 |
| 46 | 2642.0 | 1600.0 | 37.718058 | 46 | 3 |
| 0 | 2675.0 | 1567.0 | 36.940123 | 0 | 2 |
| 7 | 2708.0 | 1534.0 | 36.162188 | 7 | 2 |
| 27 | 2711.0 | 1531.0 | 36.091466 | 27 | 2 |
| 26 | 2713.0 | 1529.0 | 36.044319 | 26 | 4 |
| 102 | 2744.0 | 1498.0 | 35.313531 | 102 | 4 |
| 44 | 2786.0 | 1456.0 | 34.323432 | 44 | 3 |
| 60 | 2838.0 | 1404.0 | 33.097595 | 60 | 3 |
| 10 | 2845.0 | 1397.0 | 32.932579 | 10 | 4 |
| 1 | 2930.0 | 1312.0 | 30.928807 | 1 | 2 |
| 17 | 2949.0 | 1293.0 | 30.480905 | 17 | 2 |

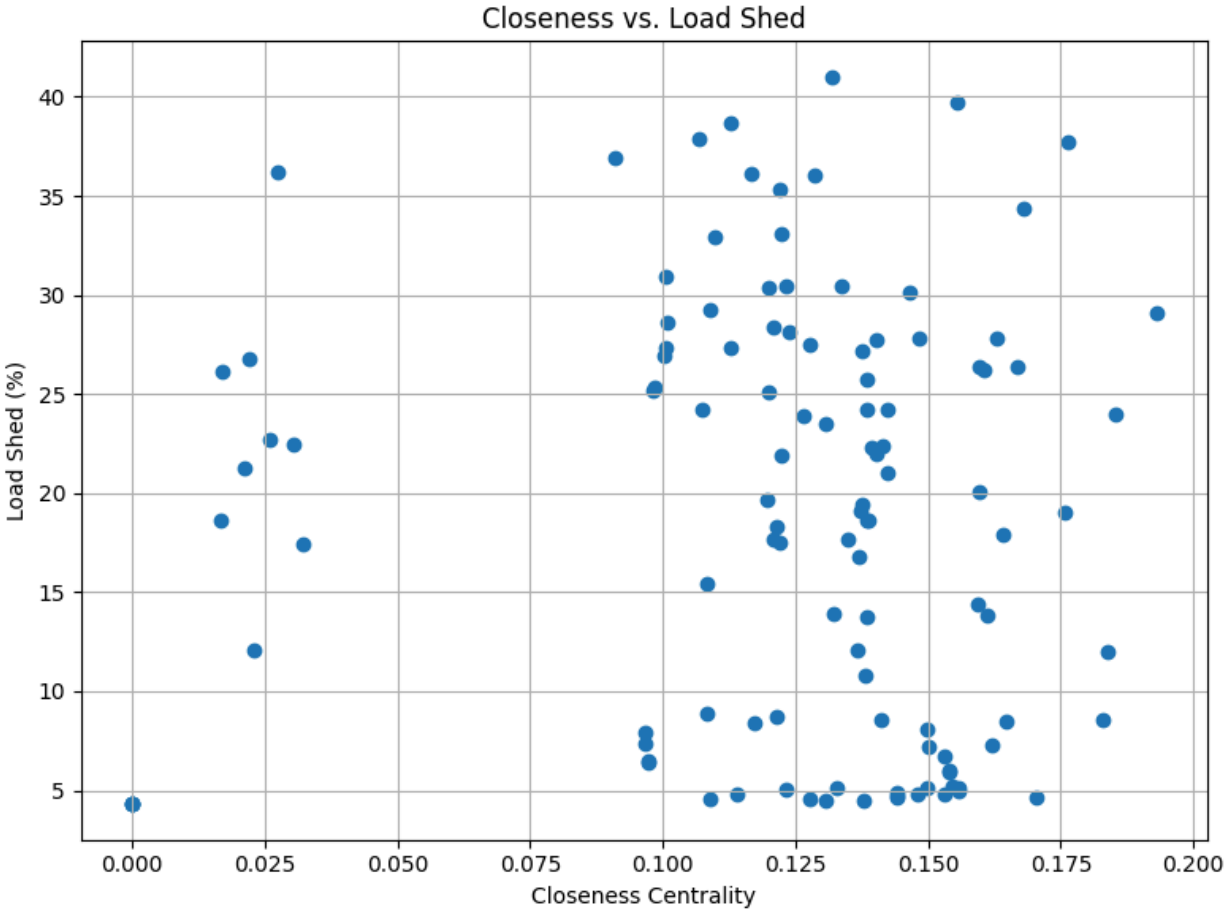
Top 15 node comparison rows

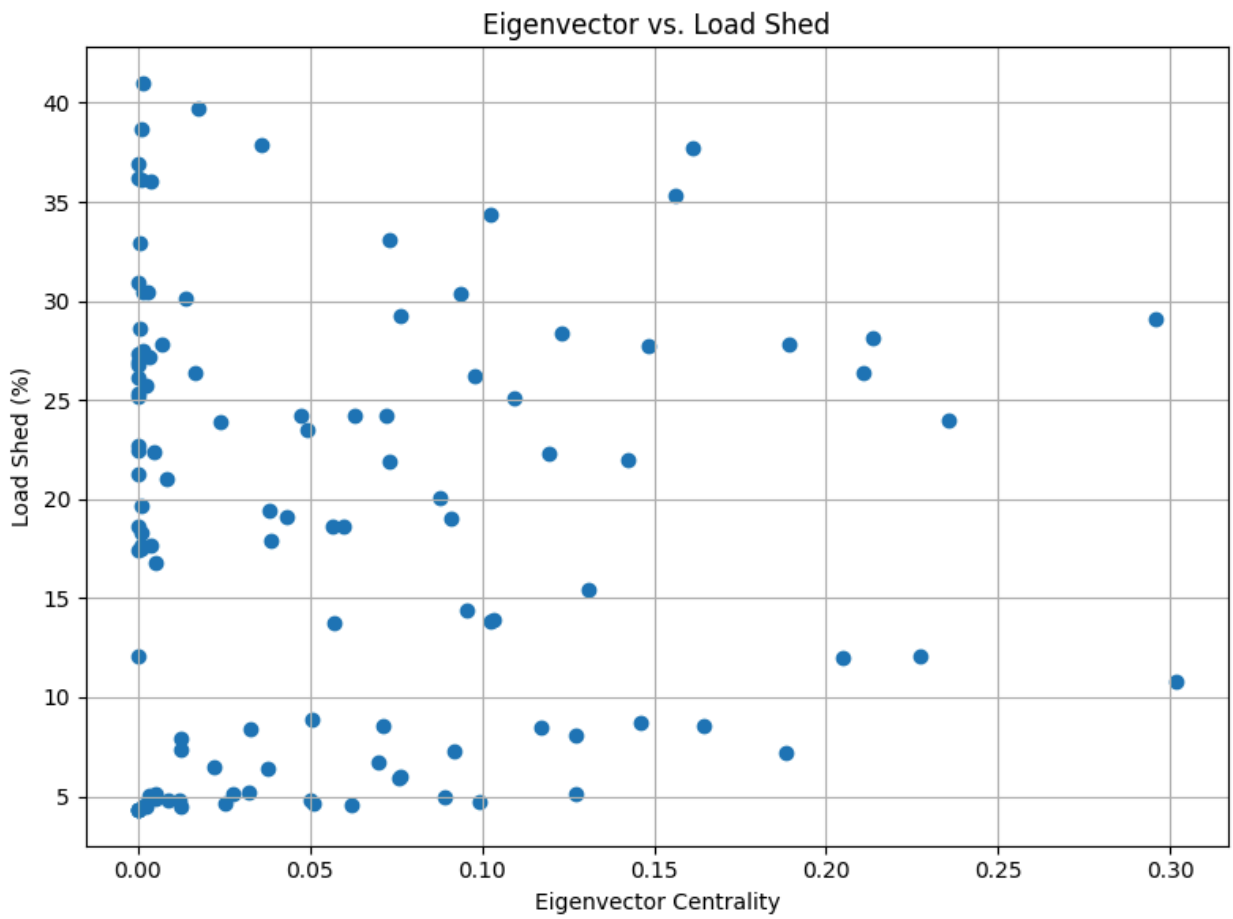
| | bus_id | degree | node_betweenness | closeness | eigenvector | initial_bus | incident_line_count | cascade_size | shed_percent | lcc_size | solver_failed |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 51 | 20 | 2 | 0.013587 | 0.131875 | 1.436099e-03 | 20 | 2 | 91 | 40.971240 | 21 | False |
| 10 | 36 | 5 | 0.139894 | 0.155369 | 1.768787e-02 | 36 | 5 | 97 | 39.745403 | 21 | False |
| 16 | 11 | 7 | 0.088705 | 0.112875 | 9.090365e-04 | 11 | 7 | 76 | 38.684583 | 32 | False |
| 88 | 89 | 2 | 0.000442 | 0.106872 | 3.579467e-02 | 89 | 3 | 97 | 37.859500 | 15 | False |
| 55 | 46 | 3 | 0.009264 | 0.176421 | 1.611995e-01 | 46 | 3 | 88 | 37.718058 | 31 | False |
| 93 | 0 | 2 | 0.000123 | 0.090899 | 1.513821e-04 | 0 | 2 | 99 | 36.940123 | 11 | False |
| 70 | 7 | 2 | 0.001768 | 0.027350 | 8.697425e-47 | 7 | 2 | 93 | 36.162188 | 31 | False |
| 87 | 27 | 2 | 0.000442 | 0.116723 | 1.112130e-03 | 27 | 2 | 74 | 36.091466 | 32 | False |
| 43 | 26 | 4 | 0.021959 | 0.128574 | 3.562076e-03 | 26 | 4 | 77 | 36.044319 | 33 | False |
| 22 | 102 | 4 | 0.067173 | 0.121958 | 1.562972e-01 | 102 | 4 | 100 | 35.313531 | 15 | False |
| 35 | 44 | 3 | 0.038901 | 0.168081 | 1.025559e-01 | 44 | 3 | 95 | 34.323432 | 15 | False |
| 82 | 60 | 3 | 0.000639 | 0.122443 | 7.293503e-02 | 60 | 3 | 70 | 33.097595 | 31 | False |
| 37 | 10 | 4 | 0.032219 | 0.109661 | 6.122798e-04 | 10 | 4 | 69 | 32.932579 | 31 | False |
| 57 | 1 | 2 | 0.007245 | 0.100483 | 2.633796e-04 | 1 | 2 | 68 | 30.928807 | 31 | False |
| 62 | 17 | 2 | 0.004605 | 0.123095 | 1.558456e-03 | 17 | 2 | 81 | 30.480905 | 32 | False |

| | metric | target | pearson | spearman |
|---|---|---|---|---|
| 0 | degree | shed_percent | 0.321864 | 0.367335 |
| 2 | node_betweenness | shed_percent | 0.101518 | 0.184004 |
| 6 | eigenvector | shed_percent | -0.028420 | -0.042670 |
| 4 | closeness | shed_percent | -0.001172 | -0.083770 |
| 1 | degree | cascade_size | 0.263728 | 0.311377 |
| 3 | node_betweenness | cascade_size | 0.076771 | 0.171700 |
| 7 | eigenvector | cascade_size | -0.065910 | -0.085887 |
| 5 | closeness | cascade_size | -0.046858 | -0.138177 |




Node Betweenness vs. Load Shed
Load Shed (%)
Node Betweenness
40
35
30
25
20
15
10
5
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35


Closeness vs. Load Shed
Load Shed (%)
Closeness Centrality
40
35
30
25
20
15
10
5
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200



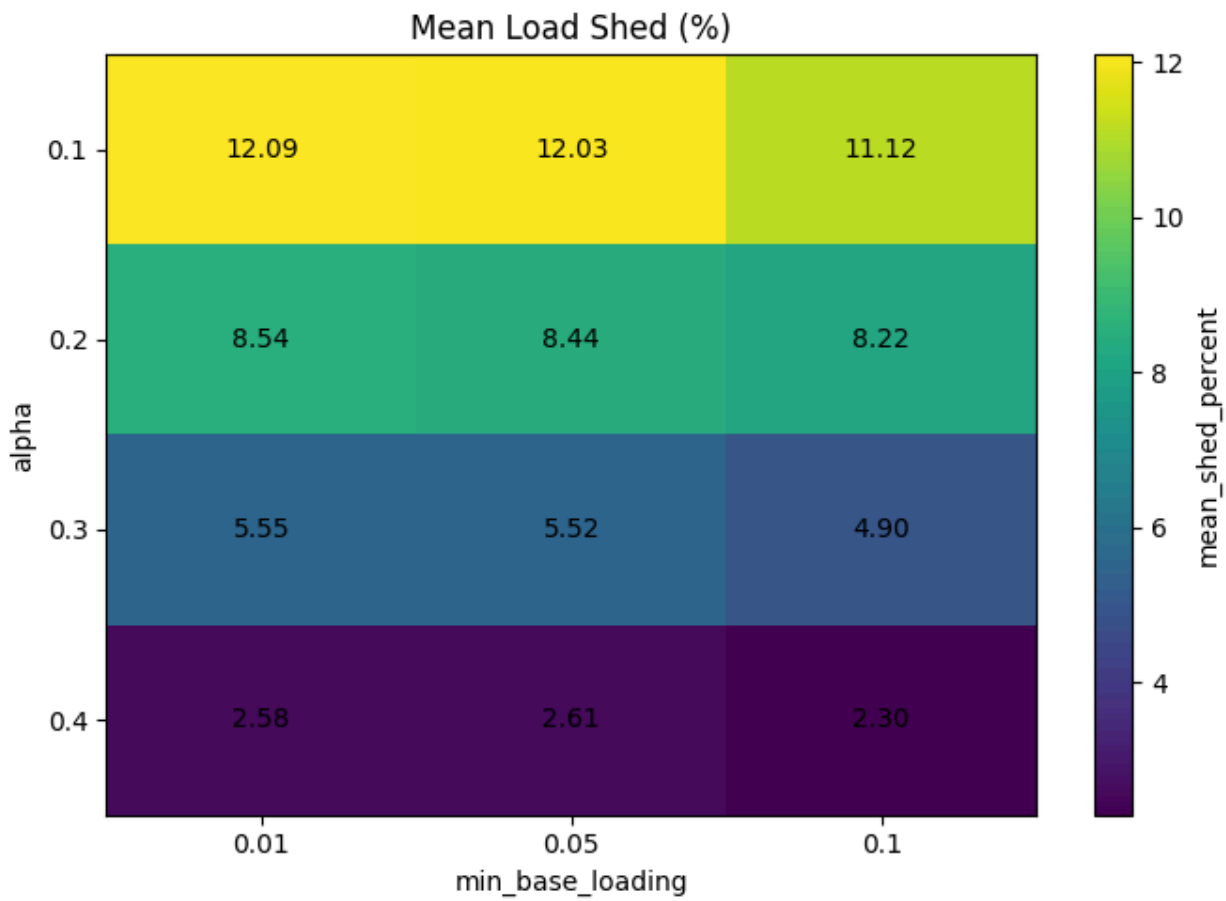

| | metric | top10_overlap_vs_shed | top10_jaccard_vs_shed | top10_overlap_vs_cascade | top10_jaccard_vs_cascade |
|---|---|---|---|---|---|
| 0 | degree | 2 | 0.111111 | 1 | 0.052632 |
| 1 | node_betweenness | 1 | 0.052632 | 2 | 0.111111 |
| 2 | closeness | 1 | 0.052632 | 2 | 0.111111 |
| 3 | eigenvector | 0 | 0.000000 | 0 | 0.000000 |

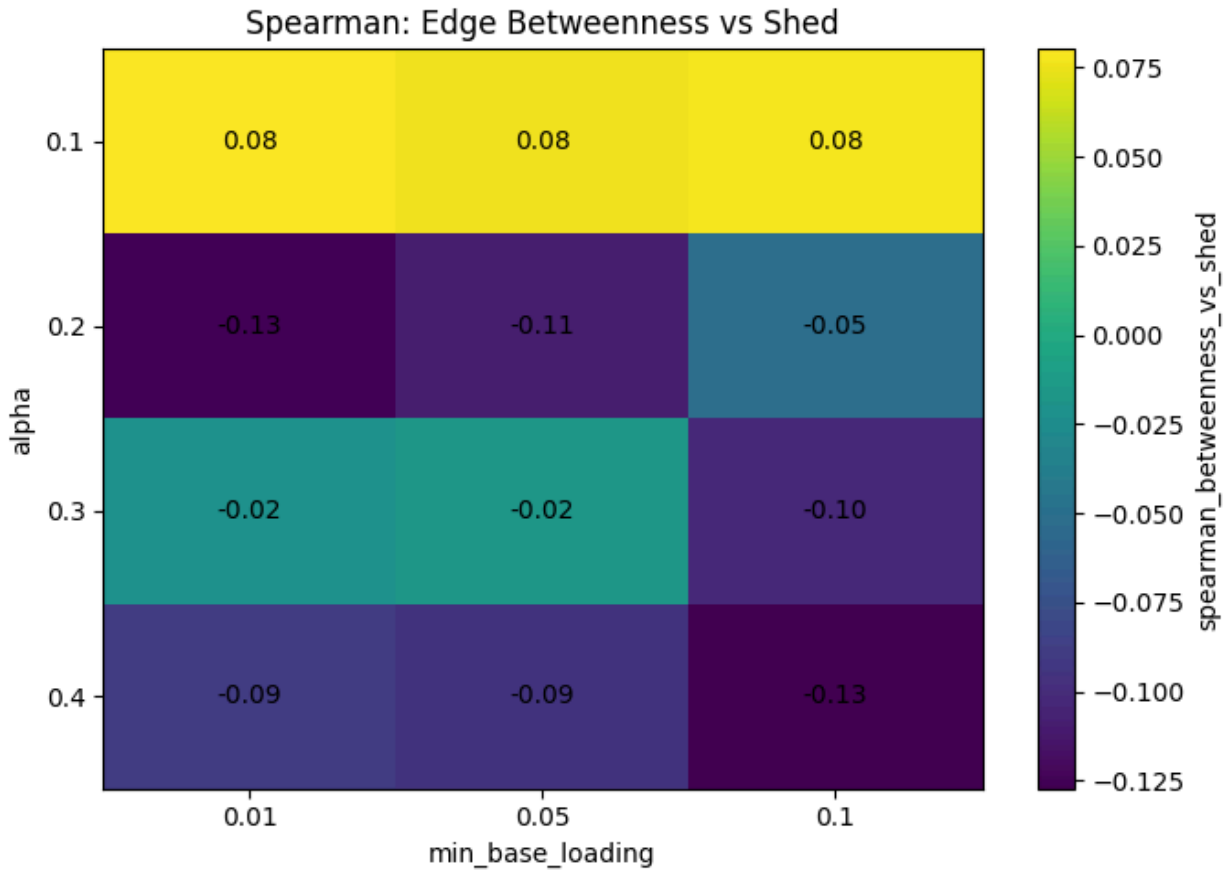
| | metric | pearson_vs_shed | spearman_vs_shed | pearson_vs_cascade | spearman_vs_cascade | top10_overlap_vs_shed | top10_jaccard_vs_shed | \ |
|---|---|---|---|---|---|---|---|---|
| 0 | degree | 0.321864 | 0.367335 | 0.263728 | 0.311377 | 2 | 0.111111 | |
| 1 | node_betweenness | 0.101518 | 0.184004 | 0.076771 | 0.171700 | 1 | 0.052632 | |
| 3 | eigenvector | -0.028420 | -0.042670 | -0.065910 | -0.085887 | 0 | 0.000000 | |
| 2 | closeness | -0.001172 | -0.083770 | -0.046858 | -0.138177 | 1 | 0.052632 | |

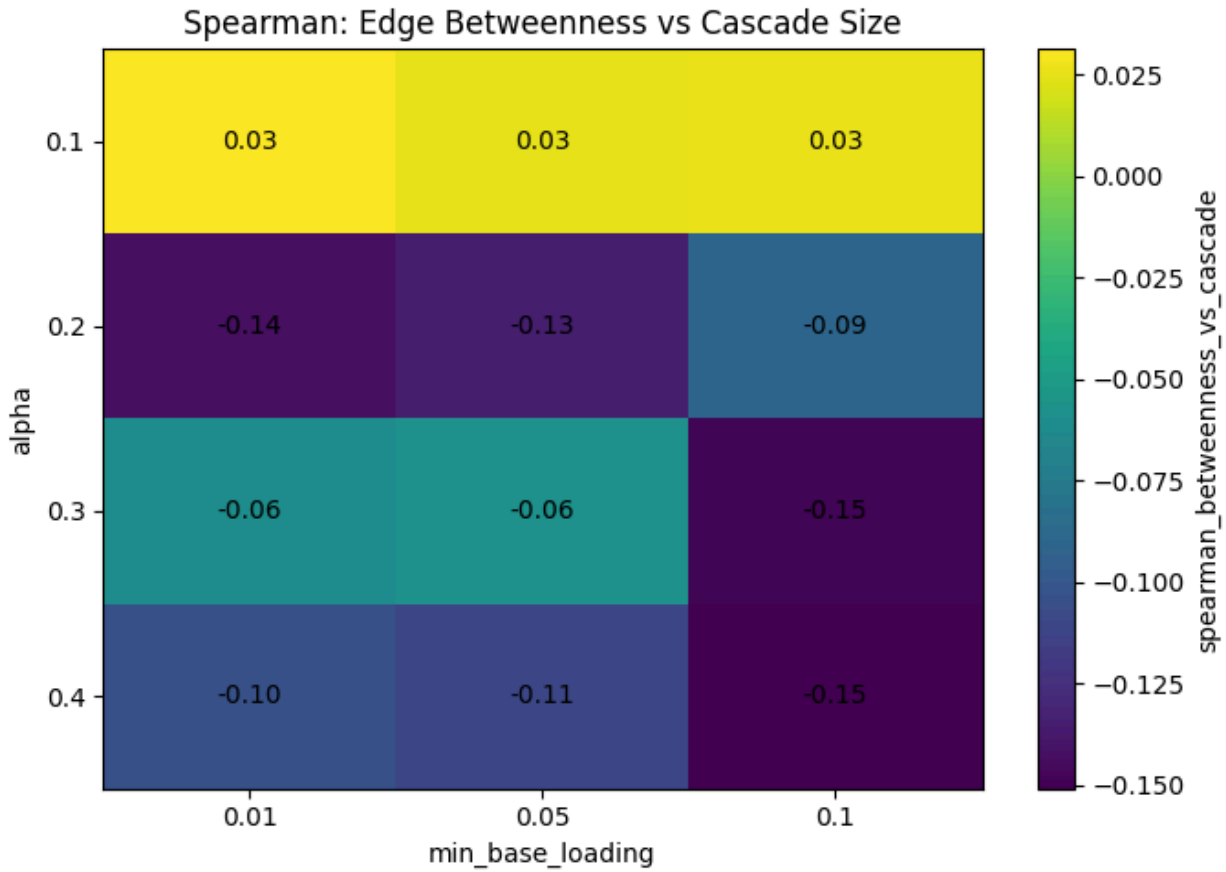
| | top10_overlap_vs_cascade | top10_jaccard_vs_cascade |
|---|---|---|
| 0 | 1 | 0.052632 |
| 1 | 2 | 0.111111 |
| 3 | 0 | 0.000000 |
| 2 | 2 | 0.111111 |

| | alpha | min_base_loading | mean_shed_percent | max_shed_percent | mean_cascade_size | max_cascade_size | spearman_betweenness_vs_shed | \ |
|---|---|---|---|---|---|---|---|---|
| 0 | 0.1 | 0.01 | 12.090627 | 24.021688 | 68.890173 | 117 | 0.080116 | |
| 1 | 0.1 | 0.05 | 12.027264 | 24.021688 | 67.976879 | 128 | 0.076196 | |
| 2 | 0.1 | 0.10 | 11.118242 | 22.960868 | 64.213873 | 128 | 0.077408 | |
| 3 | 0.2 | 0.01 | 8.538480 | 19.165488 | 50.797688 | 100 | -0.125416 | |
| 4 | 0.2 | 0.05 | 8.440369 | 18.976898 | 50.265896 | 99 | -0.109261 | |
| 5 | 0.2 | 0.10 | 8.223708 | 22.253654 | 47.138728 | 109 | -0.052588 | |
| 6 | 0.3 | 0.01 | 5.552649 | 18.929750 | 30.745665 | 103 | -0.021883 | |
| 7 | 0.3 | 0.05 | 5.518582 | 18.929750 | 30.271676 | 103 | -0.018166 | |
| 8 | 0.3 | 0.10 | 4.904983 | 19.660537 | 26.682081 | 104 | -0.102847 | |
| 10 | 0.4 | 0.05 | 2.605380 | 16.902405 | 16.849711 | 94 | -0.094641 | |
| 9 | 0.4 | 0.01 | 2.581670 | 16.902405 | 16.757225 | 94 | -0.089972 | |
| 11 | 0.4 | 0.10 | 2.302055 | 16.902405 | 15.213873 | 92 | -0.127131 | |

| | spearman_betweenness_vs_cascade | top10_overlap | top10_jaccard |
|---|---|---|---|
| 0 | 0.031174 | 2 | 0.111111 |
| 1 | 0.025122 | 2 | 0.111111 |
| 2 | 0.026254 | 0 | 0.000000 |
| 3 | -0.142726 | 0 | 0.000000 |
| 4 | -0.134885 | 0 | 0.000000 |
| 5 | -0.089691 | 0 | 0.000000 |
| 6 | -0.059189 | 0 | 0.000000 |
| 7 | -0.058313 | 0 | 0.000000 |
| 8 | -0.148578 | 0 | 0.000000 |
| 10 | -0.110267 | 2 | 0.111111 |
| 9 | -0.103942 | 2 | 0.111111 |
| 11 | -0.150854 | 2 | 0.111111 |




Spearman: Edge Betweenness vs Shed
0.08
0.08
0.08
-0.13
-0.11
-0.05
-0.02
-0.02
-0.10
-0.09
-0.09
-0.13
0.1
0.2
0.3
0.4
alpha
0.01
0.05
0.1
min_base_loading
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
−0.125
spearman_betweenness_vs_shed




Top 10 edge-triggered cascades

| | initial_line | initial_lines | cascade_size | tripped_lines | iterations | solver_failed | lcc_size | served_load_mw | \ |
|---|---|---|---|---|---|---|---|---|---|
| 19 | 19 | [19] | 99 | [19, 13, 14, 15, 16, 17, 20, 21, 22, 23, 24, 2... | 8 | False | 23 | 2323.0 | |
| 25 | 25 | [25] | 87 | [25, 22, 23, 24, 28, 40, 41, 46, 49, 99, 102, ... | 8 | False | 28 | 2518.0 | |
| 26 | 26 | [26] | 90 | [26, 22, 23, 24, 28, 166, 32, 38, 40, 41, 46, ... | 8 | False | 21 | 2536.0 | |
| 27 | 27 | [27] | 80 | [27, 21, 22, 23, 24, 28, 32, 37, 38, 41, 165, ... | 9 | False | 32 | 2536.0 | |
| 125 | 125 | [125] | 98 | [125, 126, 127, 130, 110, 116, 117, 118, 119, ... | 15 | False | 15 | 2561.0 | |
| 126 | 126 | [126] | 98 | [126, 125, 127, 130, 110, 116, 117, 118, 119, ... | 15 | False | 15 | 2561.0 | |
| 0 | 0 | [0] | 99 | [0, 1, 15, 17, 18, 28, 40, 49, 51, 14, 16, 19,... | 13 | False | 11 | 2675.0 | |
| 58 | 58 | [58] | 87 | [58, 59, 60, 63, 64, 23, 40, 41, 43, 46, 52, 5... | 6 | False | 31 | 2676.0 | |
| 64 | 64 | [64] | 87 | [64, 58, 59, 60, 63, 23, 40, 41, 43, 46, 52, 5... | 6 | False | 31 | 2676.0 | |
| 31 | 31 | [31] | 109 | [31, 35, 36, 38, 39, 168, 23, 28, 29, 40, 49, ... | 13 | False | 11 | 2694.0 | |

| | shed_load_mw | shed_percent |
|---|---|---|
| 19 | 1919.0 | 45.238095 |
| 25 | 1724.0 | 40.641207 |
| 26 | 1706.0 | 40.216879 |
| 27 | 1706.0 | 40.216879 |
| 125 | 1681.0 | 39.627534 |
| 126 | 1681.0 | 39.627534 |
| 0 | 1567.0 | 36.940123 |
| 58 | 1566.0 | 36.916549 |
| 64 | 1566.0 | 36.916549 |
| 31 | 1548.0 | 36.492221 |

Top 10 node-triggered cascades

| | initial_line | initial_lines | cascade_size | tripped_lines | iterations | solver_failed | lcc_size | \ |
|---|---|---|---|---|---|---|---|---|
| 20 | NaN | [25, 26] | 91 | [25, 26, 22, 23, 24, 28, 166, 32, 38, 40, 41, ... | 8 | False | 21 | |
| 36 | NaN | [43, 44, 46, 47, 48] | 97 | [43, 44, 46, 47, 48, 13, 14, 15, 16, 17, 19, 2... | 4 | False | 21 | |
| 11 | NaN | [10, 11, 12, 13, 15, 18, 170] | 76 | [10, 11, 12, 13, 15, 18, 170, 0, 1, 3, 17, 20,... | 10 | False | 32 | |
| 89 | NaN | [125, 126, 127] | 97 | [125, 126, 127, 110, 116, 117, 118, 119, 120, ... | 14 | False | 15 | |
| 46 | NaN | [58, 60, 96] | 88 | [58, 60, 96, 59, 63, 64, 23, 40, 41, 43, 46, 5... | 6 | False | 31 | |
| 0 | NaN | [0, 1] | 99 | [0, 1, 15, 17, 18, 28, 40, 49, 51, 14, 16, 19,... | 12 | False | 11 | |
| 7 | NaN | [6, 33] | 93 | [6, 33, 0, 5, 10, 11, 12, 13, 14, 15, 16, 17, ... | 4 | False | 31 | |
| 27 | NaN | [31, 32] | 74 | [31, 32, 36, 38, 39, 168, 23, 28, 35, 167, 169... | 11 | False | 32 | |
| 26 | NaN | [30, 31, 39, 168] | 77 | [30, 31, 39, 168, 23, 25, 28, 29, 34, 35, 36, ... | 5 | False | 33 | |
| 102 | NaN | [150, 152, 153, 161] | 100 | [150, 152, 153, 161, 139, 140, 151, 154, 155, ... | 13 | False | 15 | |

| | served_load_mw | shed_load_mw | shed_percent | initial_bus | incident_line_count |
|---|---|---|---|---|---|
| 20 | 2504.0 | 1738.0 | 40.971240 | 20 | 2 |
| 36 | 2556.0 | 1686.0 | 39.745403 | 36 | 5 |
| 11 | 2601.0 | 1641.0 | 38.684583 | 11 | 7 |
| 89 | 2636.0 | 1606.0 | 37.859500 | 89 | 3 |
| 46 | 2642.0 | 1600.0 | 37.718058 | 46 | 3 |
| 0 | 2675.0 | 1567.0 | 36.940123 | 0 | 2 |
| 7 | 2708.0 | 1534.0 | 36.162188 | 7 | 2 |
| 27 | 2711.0 | 1531.0 | 36.091466 | 27 | 2 |
| 26 | 2713.0 | 1529.0 | 36.044319 | 26 | 4 |
| 102 | 2744.0 | 1498.0 | 35.313531 | 102 | 4 |

Node metric summary

| | metric | pearson_vs_shed | spearman_vs_shed | pearson_vs_cascade | spearman_vs_cascade | top10_overlap_vs_shed | top10_jaccard_vs_shed | \ |
|---|---|---|---|---|---|---|---|---|
| 0 | degree | 0.321864 | 0.367335 | 0.263728 | 0.311377 | 2 | 0.111111 | |
| 1 | node_betweenness | 0.101518 | 0.184004 | 0.076771 | 0.171700 | 1 | 0.052632 | |
| 3 | eigenvector | -0.028420 | -0.042670 | -0.065910 | -0.085887 | 0 | 0.000000 | |
| 2 | closeness | -0.001172 | -0.083770 | -0.046858 | -0.138177 | 1 | 0.052632 | |

| | top10_overlap_vs_cascade | top10_jaccard_vs_cascade |
|---|---|---|
| 0 | 1 | 0.052632 |
| 1 | 2 | 0.111111 |
| 3 | 0 | 0.000000 |
| 2 | 2 | 0.111111 |

Case300 Results:

```
Buses: 300
Lines: 283
Loads: 193
Generators: 68
Baseline max line loading (%): 8.732496523275994
```

```
Nodes: 300
Edges: 281
```

```
Top 10 nodes by node betweenness
    bus_id  degree  node_betweenness  closeness  eigenvector
40      40       3          0.035092   0.053755     0.016229
60      60       4          0.033400   0.049786     0.006374
37      37       3          0.026309   0.054250     0.047355
30      30       7          0.025040   0.053031     0.503937
14      14       4          0.024497   0.049267     0.263022
13      13       2          0.023232   0.044358     0.073829
7        7       3          0.022536   0.040202     0.022459
36      36       4          0.021631   0.054881     0.137256
10      10       3          0.018563   0.036418     0.006790
65      65       4          0.018507   0.044694     0.003064

Top 10 edges by edge betweenness
    line_id  from_bus  to_bus  edge_betweenness
61       73        40      60          0.033445
56       68        37      40          0.025739
16       28        13      14          0.024615
11       23         7      13          0.023411
53       65        36      37          0.019899
10       22         7      10          0.019532
87       99        60      65          0.019365
47       59        31      36          0.018374
17       29        14      30          0.018131
40       52        30      31          0.016090
```

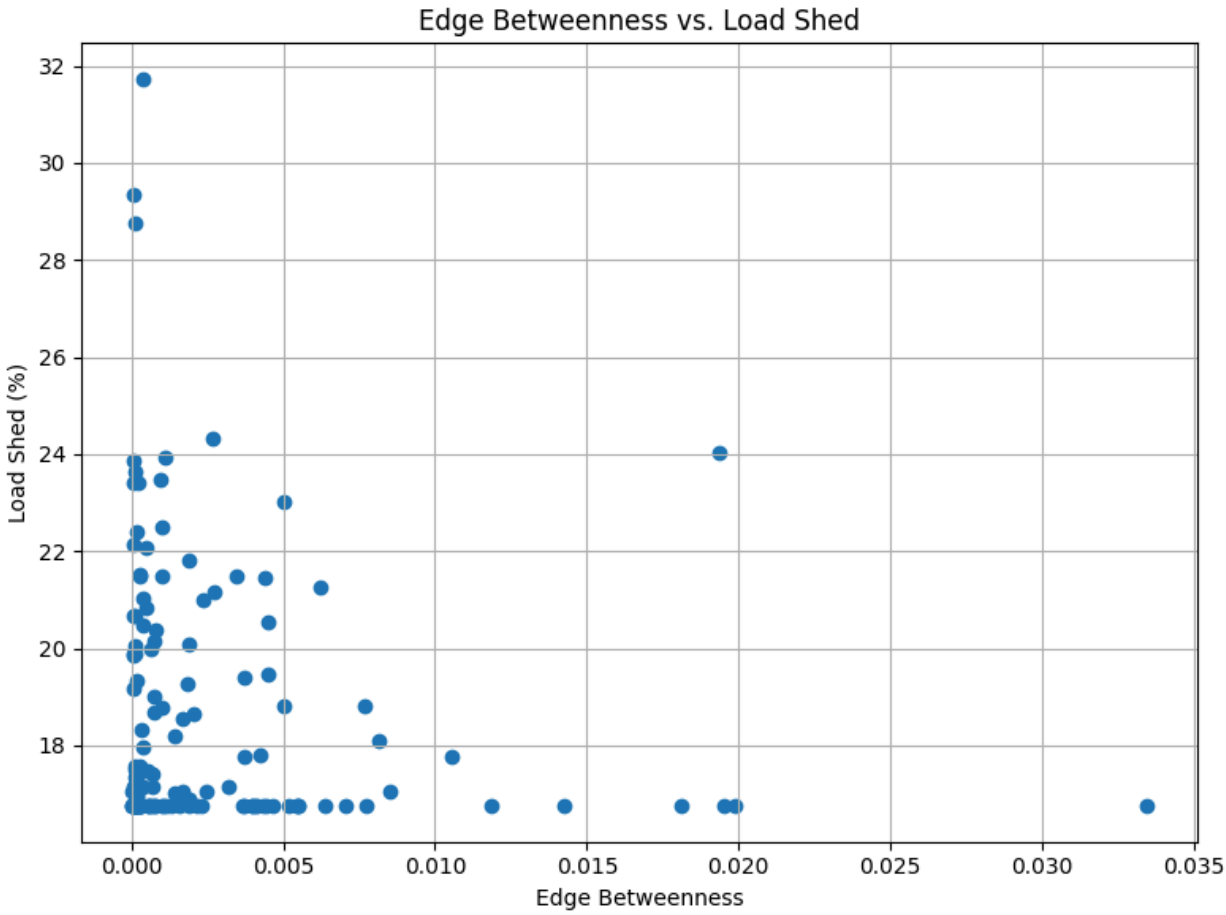
| | initial_line | initial_lines | cascade_size | tripped_lines | iterations | solver_failed | lcc_size | served_load_mw | \ |
|---|---|---|---|---|---|---|---|---|---|
| 225 | 225 | [225] | 79 | [225, 179, 181, 220, 221, 0, 14, 16, 17, 18, 1... | 3 | False | 30 | 16283.52 | |
| 201 | 201 | [201] | 64 | [201, 145, 149, 150, 151, 153, 156, 157, 202, ... | 3 | False | 28 | 16847.52 | |
| 147 | 147 | [147] | 66 | [147, 145, 146, 149, 150, 151, 152, 153, 156, ... | 3 | False | 28 | 16987.82 | |
| 228 | 228 | [228] | 69 | [228, 254, 261, 262, 267, 268, 269, 29, 30, 31... | 4 | True | 34 | 17583.22 | |
| 269 | 269 | [269] | 74 | [269, 228, 257, 258, 259, 261, 263, 268, 29, 3... | 4 | True | 34 | 17586.22 | |
| 263 | 263 | [263] | 42 | [263, 228, 258, 260, 261, 265, 268, 269, 270, ... | 3 | True | 68 | 18007.22 | |
| 177 | 177 | [177] | 74 | [177, 57, 63, 159, 161, 176, 179, 196, 197, 19... | 3 | False | 25 | 18046.32 | |
| 99 | 99 | [99] | 81 | [99, 50, 91, 96, 102, 48, 49, 51, 65, 73, 90, ... | 7 | False | 34 | 18118.72 | |
| 262 | 262 | [262] | 46 | [262, 257, 258, 268, 270, 271, 0, 46, 52, 53, ... | 3 | False | 40 | 18144.42 | |
| 214 | 214 | [214] | 36 | [214, 213, 216, 217, 218, 17, 21, 52, 53, 55, ... | 3 | False | 82 | 18159.02 | |

| | shed_load_mw | shed_percent |
|---|---|---|
| 225 | 7564.13 | 31.718555 |
| 201 | 7000.13 | 29.353542 |
| 147 | 6859.83 | 28.765224 |
| 228 | 6264.43 | 26.268542 |
| 269 | 6261.43 | 26.255962 |
| 263 | 5840.43 | 24.490589 |
| 177 | 5801.33 | 24.326632 |
| 99 | 5728.93 | 24.023038 |
| 262 | 5703.23 | 23.915270 |
| 214 | 5688.63 | 23.854049 |

| | pearson_betweenness_vs_shed | spearman_betweenness_vs_shed | pearson_betweenness_vs_cascade | spearman_betweenness_vs_cascade |
|---|---|---|---|---|
| 0 | -0.090316 | -0.045879 | -0.060788 | -0.021118 |

Top 15 edge comparison rows

| | line_id | from_bus | to_bus | edge_betweenness | initial_line | cascade_size | shed_percent | lcc_size | solver_failed |
|---|---|---|---|---|---|---|---|---|---|
| 171 | 225 | 116 | 166 | 0.000386 | 225 | 79 | 31.718555 | 30 | False |
| 264 | 201 | 135 | 137 | 0.000067 | 201 | 64 | 29.353542 | 28 | False |
| 233 | 147 | 95 | 96 | 0.000134 | 147 | 66 | 28.765224 | 28 | False |
| 189 | 228 | 168 | 209 | 0.000290 | 228 | 69 | 26.268542 | 34 | True |
| 158 | 269 | 209 | 215 | 0.000535 | 269 | 74 | 26.255962 | 34 | True |
| 179 | 263 | 199 | 201 | 0.000312 | 263 | 42 | 24.490589 | 68 | True |
| 67 | 177 | 114 | 130 | 0.002676 | 177 | 74 | 24.326632 | 25 | False |
| 6 | 99 | 60 | 65 | 0.019365 | 99 | 81 | 24.023038 | 34 | False |
| 111 | 262 | 198 | 216 | 0.001115 | 262 | 46 | 23.915270 | 40 | False |
| 266 | 214 | 153 | 154 | 0.000067 | 214 | 36 | 23.854049 | 82 | False |
| 263 | 203 | 136 | 138 | 0.000067 | 203 | 51 | 23.633901 | 59 | True |
| 242 | 156 | 101 | 102 | 0.000111 | 156 | 85 | 23.631385 | 25 | False |
| 121 | 271 | 213 | 216 | 0.000981 | 271 | 21 | 23.487555 | 57 | False |
| 252 | 206 | 141 | 142 | 0.000089 | 206 | 48 | 23.414592 | 71 | False |
| 206 | 20 | 6 | 11 | 0.000223 | 20 | 71 | 23.402432 | 34 | False |



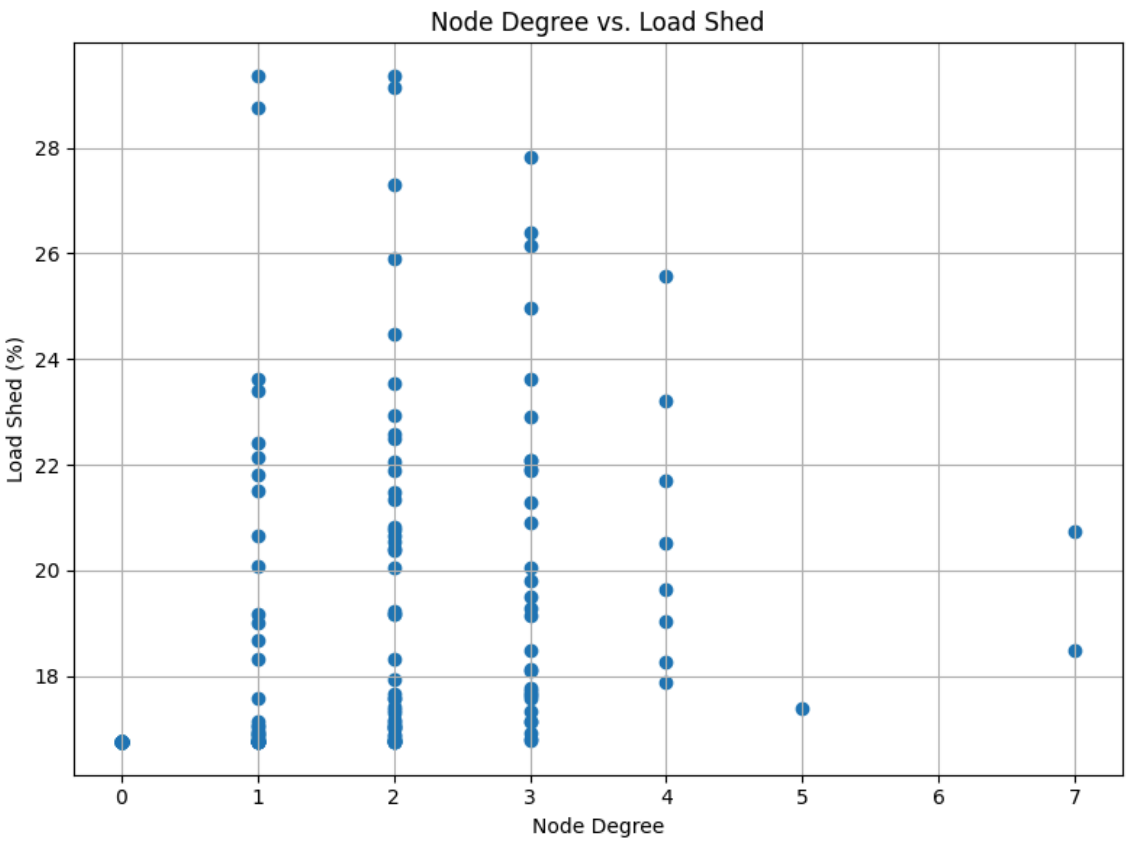

| | initial_line | initial_lines | cascade_size | tripped_lines | iterations | solver_failed | lcc_size | served_load_mw | \ |
|---|---|---|---|---|---|---|---|---|---|
| 137 | NaN | [201, 202] | 67 | [201, 202, 145, 146, 149, 150, 151, 153, 155, ... | 3 | False | 28 | 16847.52 | |
| 135 | 201.0 | [201] | 64 | [201, 145, 149, 150, 151, 153, 156, 157, 202, ... | 3 | False | 28 | 16847.52 | |
| 116 | NaN | [220, 225] | 73 | [220, 225, 0, 14, 16, 17, 18, 19, 20, 21, 22, ... | 2 | False | 44 | 16899.82 | |
| 95 | 147.0 | [147] | 66 | [147, 145, 146, 149, 150, 151, 152, 153, 156, ... | 3 | False | 28 | 16987.82 | |
| 18 | NaN | [27, 35, 36] | 70 | [27, 35, 36, 20, 22, 39, 41, 42, 16, 29, 30, 3... | 4 | False | 34 | 17214.32 | |
| 215 | NaN | [259, 269] | 64 | [259, 269, 98, 228, 254, 258, 261, 262, 263, 2... | 3 | False | 39 | 17335.22 | |
| 114 | NaN | [175, 176, 177] | 85 | [175, 176, 177, 159, 161, 170, 173, 174, 179, ... | 3 | False | 25 | 17553.02 | |
| 168 | 228.0 | [228] | 69 | [228, 254, 261, 262, 267, 268, 269, 29, 30, 31... | 4 | True | 34 | 17583.22 | |
| 6 | NaN | [15, 20, 21] | 75 | [15, 20, 21, 13, 16, 17, 19, 22, 24, 25, 27, 1... | 4 | False | 37 | 17615.22 | |
| 98 | NaN | [146, 150, 152] | 58 | [146, 150, 152, 0, 17, 18, 20, 21, 22, 23, 24,... | 2 | True | 59 | 17621.62 | |
| 209 | NaN | [228, 269] | 48 | [228, 269, 65, 257, 258, 259, 260, 261, 262, 2... | 3 | False | 43 | 17671.42 | |
| 96 | NaN | [147, 148, 149, 151] | 66 | [147, 148, 149, 151, 145, 146, 152, 153, 156, ... | 3 | False | 28 | 17750.72 | |
| 153 | NaN | [212, 214, 215] | 40 | [212, 214, 215, 213, 216, 217, 218, 17, 21, 46... | 3 | False | 71 | 17891.82 | |
| 201 | 263.0 | [263] | 42 | [263, 228, 258, 260, 261, 265, 268, 269, 270, ... | 3 | True | 68 | 18007.22 | |
| 216 | NaN | [262, 271] | 64 | [262, 271, 0, 46, 52, 53, 54, 55, 56, 57, 58, ... | 2 | False | 34 | 18013.42 | |

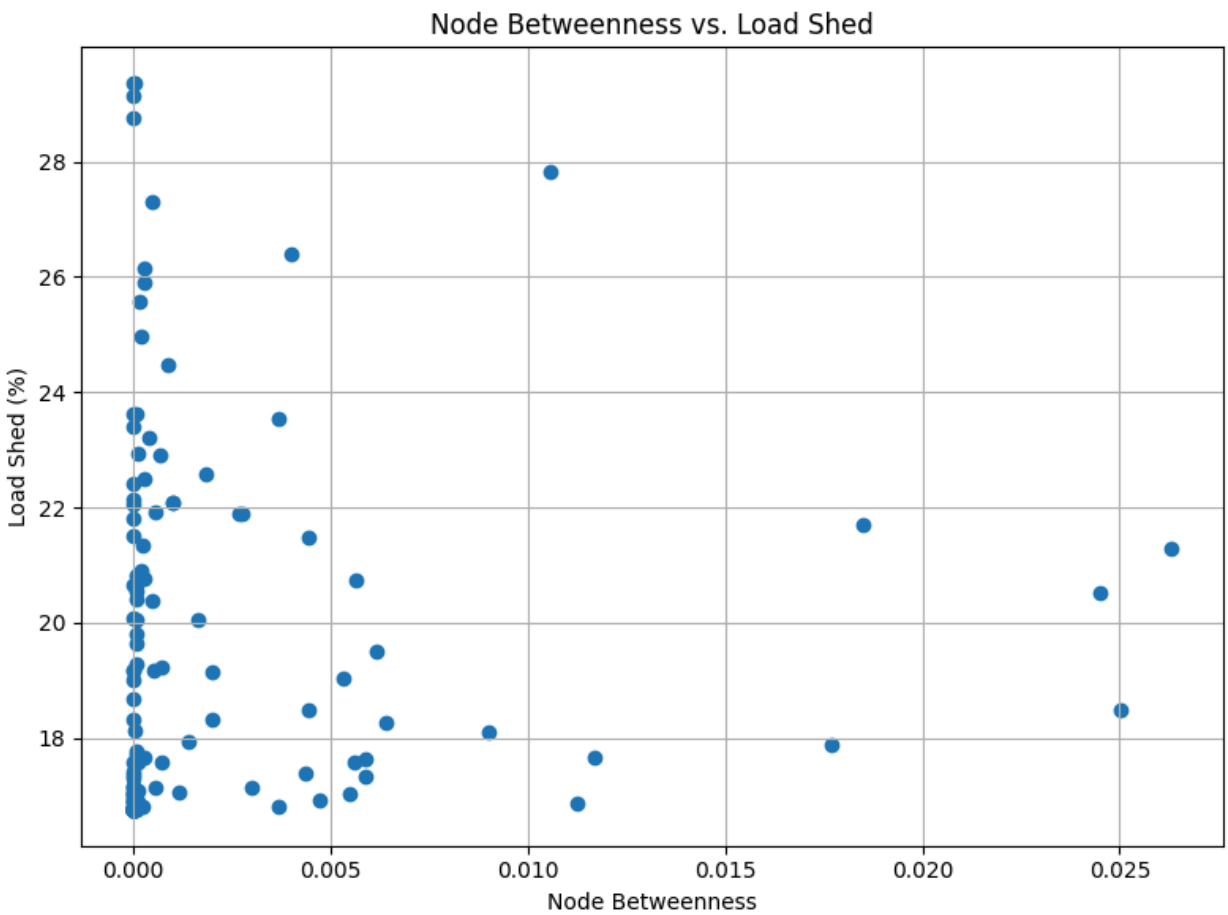
| | shed_load_mw | shed_percent | initial_bus | incident_line_count |
|---|---|---|---|---|
| 137 | 7000.13 | 29.353542 | 137 | 2 |
| 135 | 7000.13 | 29.353542 | 135 | 1 |
| 116 | 6947.83 | 29.134233 | 116 | 2 |
| 95 | 6859.83 | 28.765224 | 95 | 1 |
| 18 | 6633.33 | 27.815445 | 18 | 3 |
| 215 | 6512.43 | 27.308477 | 215 | 2 |
| 114 | 6294.63 | 26.395179 | 114 | 3 |
| 168 | 6264.43 | 26.268542 | 168 | 1 |
| 6 | 6232.43 | 26.134357 | 6 | 3 |
| 98 | 6226.03 | 26.107520 | 98 | 3 |
| 209 | 6176.23 | 25.898694 | 209 | 2 |
| 96 | 6096.93 | 25.566167 | 96 | 4 |
| 153 | 5955.83 | 24.974494 | 153 | 3 |
| 201 | 5840.43 | 24.490589 | 201 | 1 |
| 216 | 5834.23 | 24.464591 | 216 | 2 |

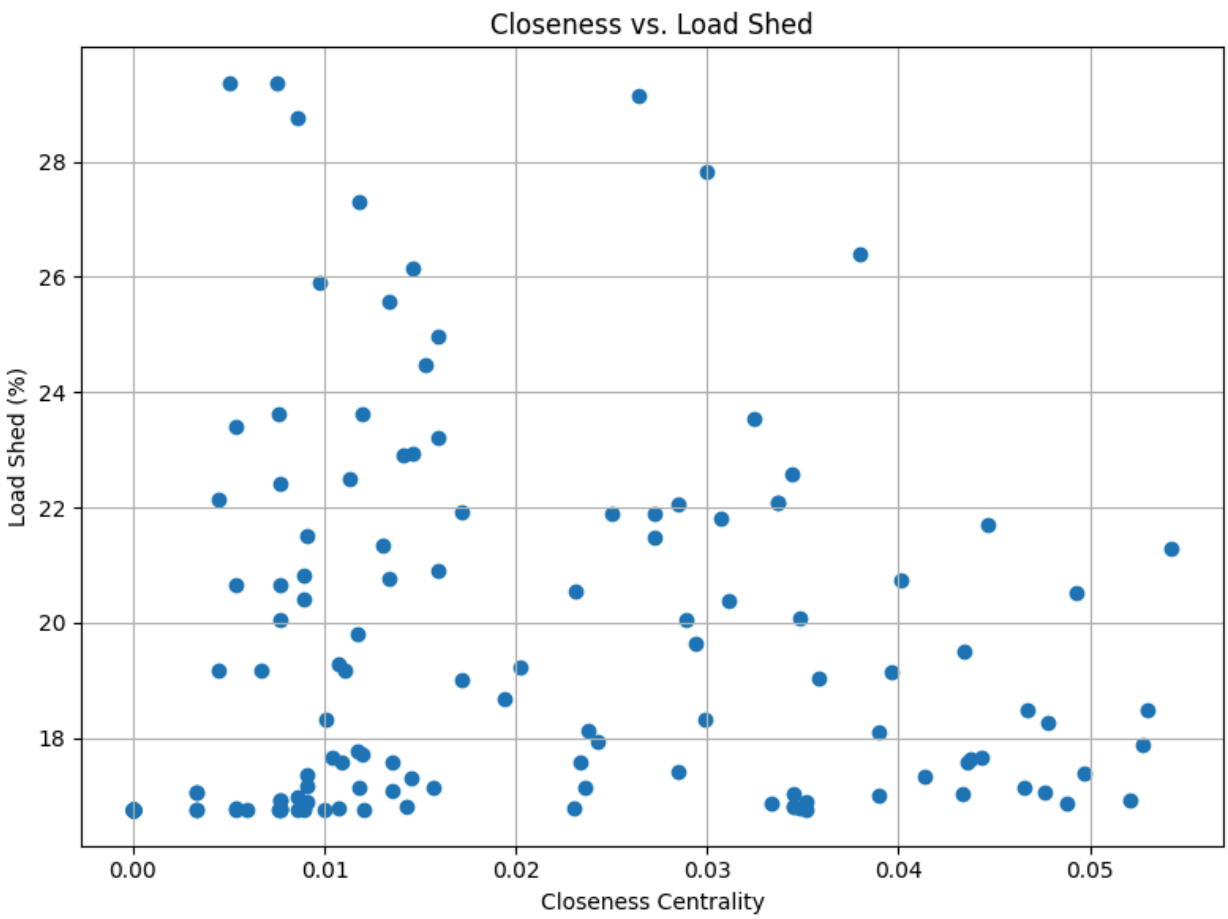
Top 15 node comparison rows

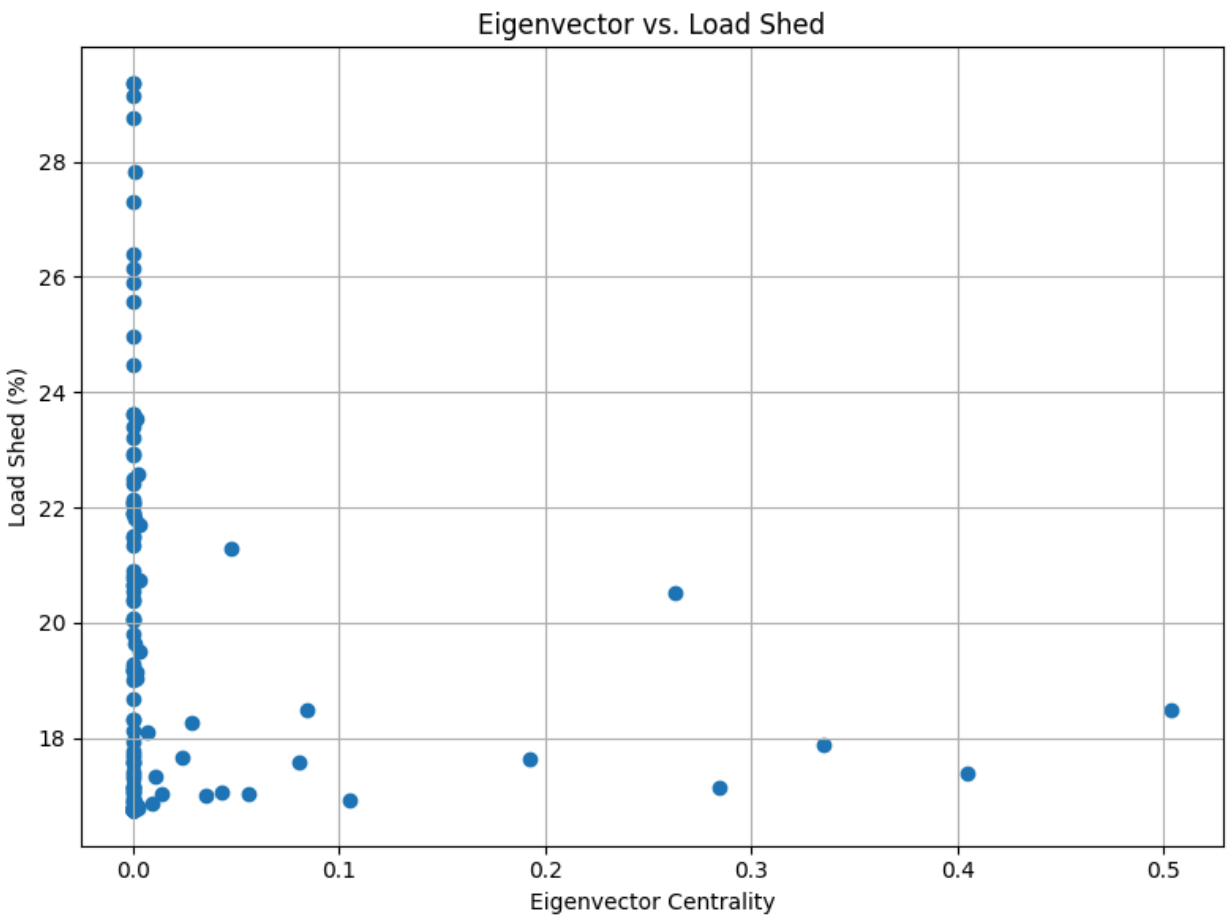
| | bus_id | degree | node_betweenness | closeness | eigenvector | initial_bus | incident_line_count | cascade_size | shed_percent | lcc_size | solver_failed |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 155 | 137 | 2 | 0.000045 | 0.007525 | 1.649029e-39 | 137 | 2 | 67 | 29.353542 | 28 | False |
| 230 | 135 | 1 | 0.000000 | 0.005017 | 1.019156e-39 | 135 | 1 | 64 | 29.353542 | 28 | False |
| 161 | 116 | 2 | 0.000007 | 0.026392 | 1.255211e-04 | 116 | 2 | 73 | 29.134233 | 44 | False |
| 243 | 95 | 1 | 0.000000 | 0.008600 | 1.794003e-19 | 95 | 1 | 66 | 28.765224 | 28 | False |
| 16 | 18 | 3 | 0.010561 | 0.029948 | 5.765319e-04 | 18 | 3 | 70 | 27.815445 | 34 | False |
| 93 | 215 | 2 | 0.000494 | 0.011775 | 2.857517e-23 | 215 | 2 | 64 | 27.308477 | 39 | False |
| 41 | 114 | 3 | 0.004025 | 0.037939 | 1.604529e-04 | 114 | 3 | 85 | 26.395179 | 25 | False |
| 107 | 6 | 3 | 0.000292 | 0.014594 | 2.214040e-23 | 6 | 3 | 75 | 26.134357 | 37 | False |
| 113 | 209 | 2 | 0.000269 | 0.009745 | 1.424607e-23 | 209 | 2 | 48 | 25.898694 | 43 | False |
| 127 | 96 | 4 | 0.000146 | 0.013378 | 4.744325e-19 | 96 | 4 | 66 | 25.566167 | 28 | False |
| 120 | 153 | 3 | 0.000202 | 0.015935 | 1.325988e-16 | 153 | 3 | 40 | 24.974494 | 71 | False |
| 77 | 216 | 2 | 0.000898 | 0.015245 | 4.356833e-21 | 216 | 2 | 64 | 24.464591 | 34 | False |
| 136 | 101 | 3 | 0.000090 | 0.011945 | 4.029683e-23 | 101 | 3 | 85 | 23.631385 | 25 | False |
| 185 | 102 | 1 | 0.000000 | 0.007601 | 1.669149e-23 | 102 | 1 | 85 | 23.631385 | 25 | False |
| 45 | 8 | 2 | 0.003681 | 0.032460 | 1.891529e-03 | 8 | 2 | 89 | 23.532004 | 30 | False |

| | metric | target | pearson | spearman |
|---|---|---|---|---|
| 0 | degree | shed_percent | 0.405637 | 0.747169 |
| 4 | closeness | shed_percent | 0.226542 | 0.676267 |
| 6 | eigenvector | shed_percent | -0.046565 | 0.658854 |
| 2 | node_betweenness | shed_percent | 0.123619 | 0.626020 |
| 1 | degree | cascade_size | 0.545163 | 0.793540 |
| 5 | closeness | cascade_size | 0.395130 | 0.716611 |
| 7 | eigenvector | cascade_size | 0.048297 | 0.700899 |
| 3 | node_betweenness | cascade_size | 0.282330 | 0.648111 |


Node Degree vs. Load Shed
Load Shed (%)
28
26
24
22
20
18
0
1
2
3
4
5
6
7
Node Degree


Node Betweenness vs. Load Shed
Load Shed (%)
28
26
24
22
20
18
0.000
0.005
0.010
0.015
0.020
0.025
Node Betweenness


Closeness vs. Load Shed
Load Shed (%)
28
26
24
22
20
18
0.00
0.01
0.02
0.03
0.04
0.05
Closeness Centrality



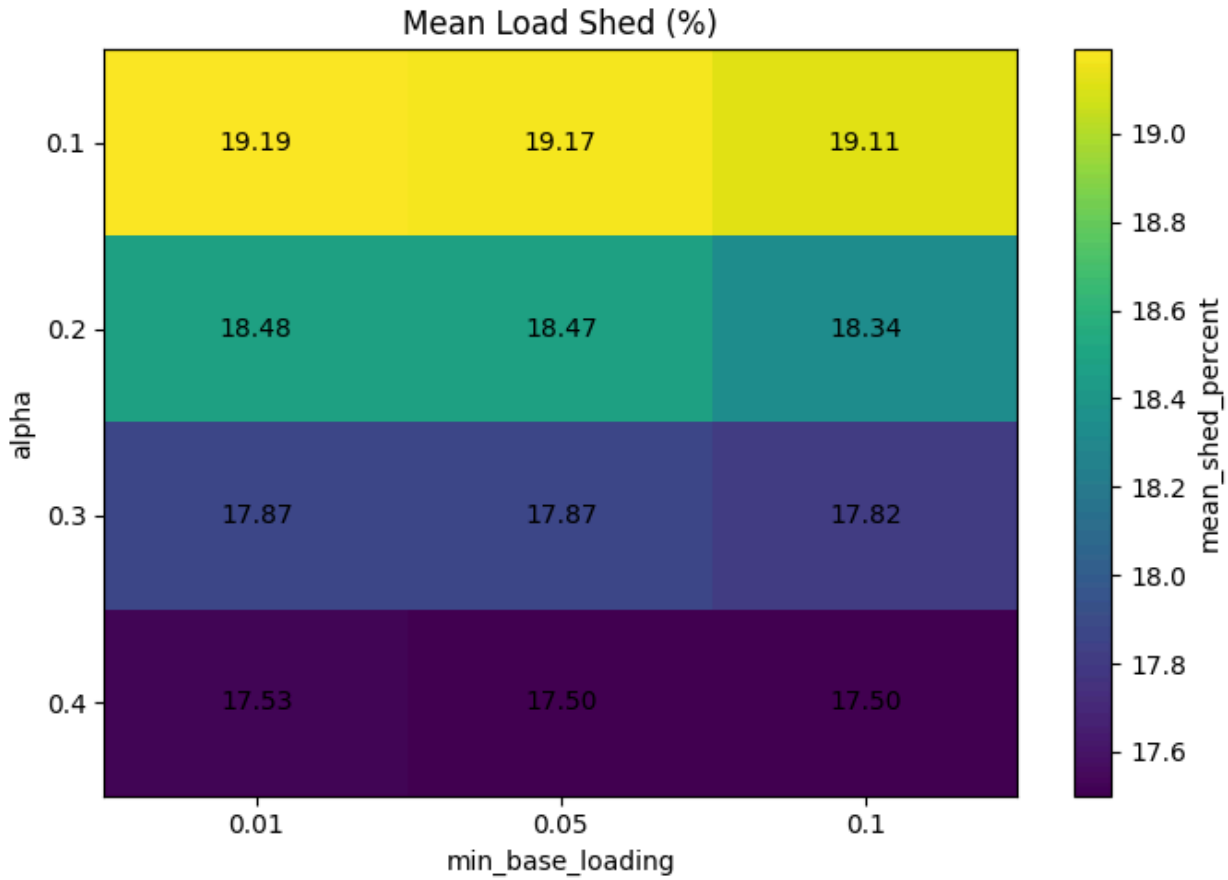

| | metric | top10_overlap_vs_shed | top10_jaccard_vs_shed | top10_overlap_vs_cascade | top10_jaccard_vs_cascade |
|---|---|---|---|---|---|
| 0 | degree | 1 | 0.052632 | 0 | 0.000000 |
| 1 | node_betweenness | 1 | 0.052632 | 1 | 0.052632 |
| 2 | closeness | 0 | 0.000000 | 0 | 0.000000 |
| 3 | eigenvector | 0 | 0.000000 | 0 | 0.000000 |

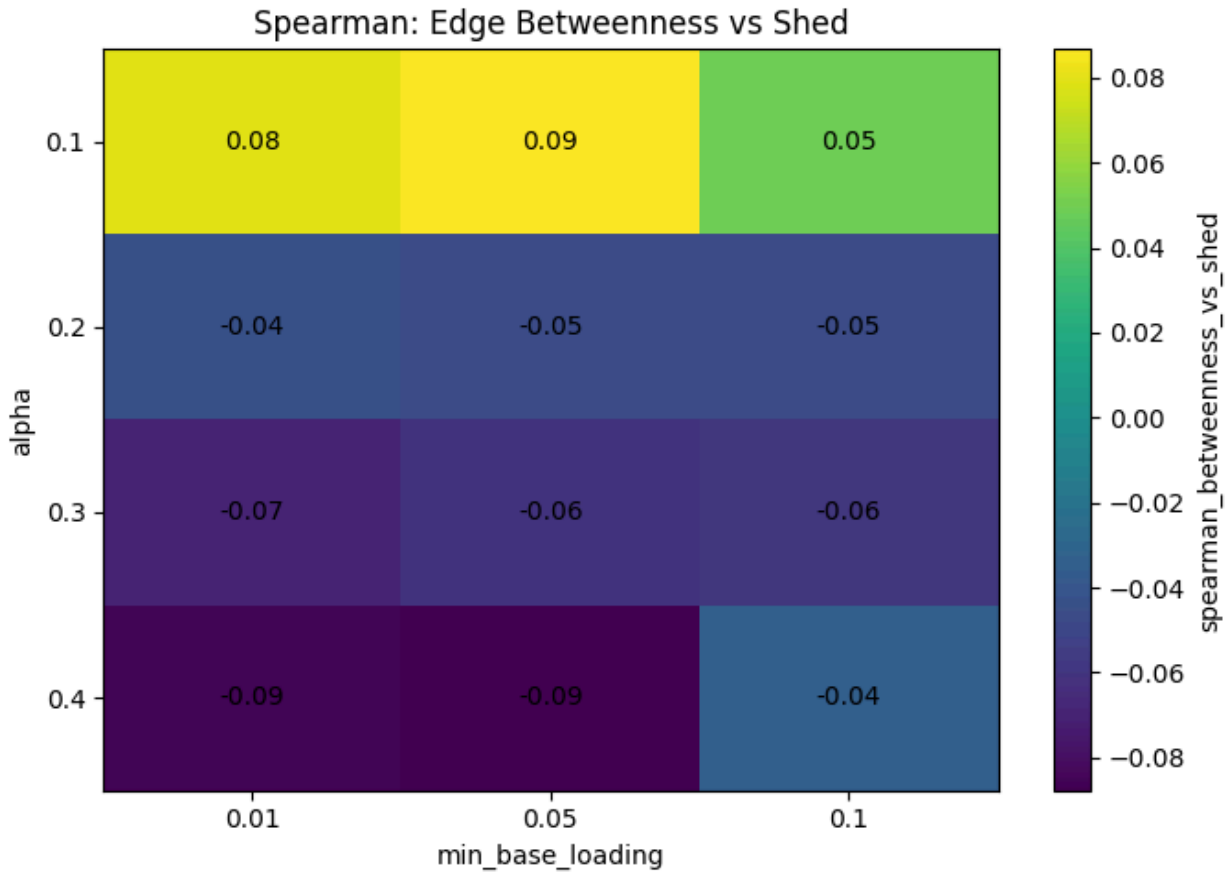
| | metric | pearson_vs_shed | spearman_vs_shed | pearson_vs_cascade | spearman_vs_cascade | top10_overlap_vs_shed | top10_jaccard_vs_shed | \ |
|---|---|---|---|---|---|---|---|---|
| 0 | degree | 0.405637 | 0.747169 | 0.545163 | 0.793540 | 1 | 0.052632 | |
| 2 | closeness | 0.226542 | 0.676267 | 0.395130 | 0.716611 | 0 | 0.000000 | |
| 3 | eigenvector | -0.046565 | 0.658854 | 0.048297 | 0.700899 | 0 | 0.000000 | |
| 1 | node_betweenness | 0.123619 | 0.626020 | 0.282330 | 0.648111 | 1 | 0.052632 | |

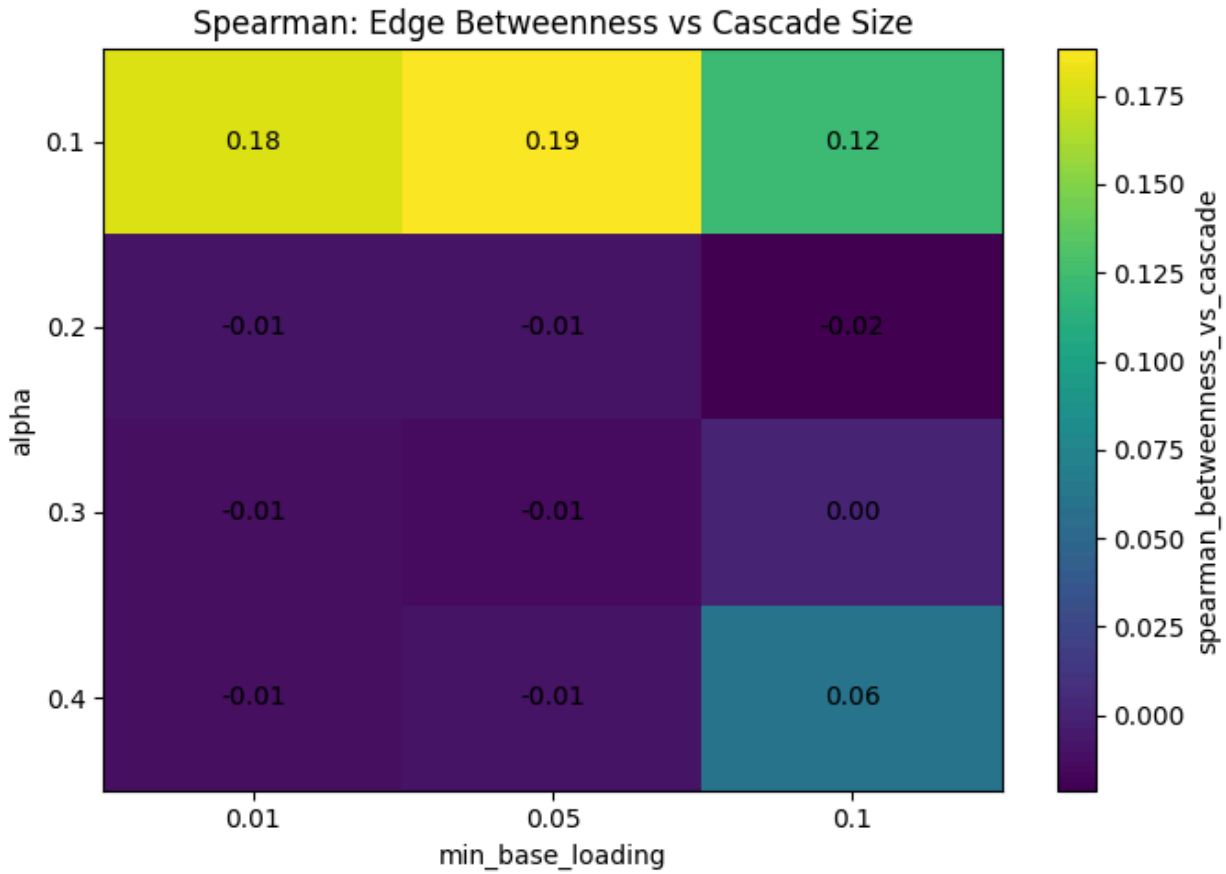
| | top10_overlap_vs_cascade | top10_jaccard_vs_cascade |
|---|---|---|
| 0 | 0 | 0.000000 |
| 2 | 0 | 0.000000 |
| 3 | 0 | 0.000000 |
| 1 | 1 | 0.052632 |

| | alpha | min_base_loading | mean_shed_percent | max_shed_percent | mean_cascade_size | max_cascade_size | spearman_betweenness_vs_shed | \ |
|---|---|---|---|---|---|---|---|---|
| 0 | 0.1 | 0.01 | 19.190539 | 30.599367 | 33.950530 | 119 | 0.078549 | |
| 1 | 0.1 | 0.05 | 19.167049 | 30.598948 | 33.395760 | 119 | 0.086641 | |
| 2 | 0.1 | 0.10 | 19.113093 | 30.545274 | 31.250883 | 128 | 0.048557 | |
| 3 | 0.2 | 0.01 | 18.484722 | 31.812485 | 23.282686 | 125 | -0.044046 | |
| 4 | 0.2 | 0.05 | 18.473767 | 31.812485 | 22.911661 | 124 | -0.045472 | |
| 5 | 0.2 | 0.10 | 18.337076 | 31.718555 | 20.515901 | 85 | -0.045879 | |
| 6 | 0.3 | 0.01 | 17.872066 | 26.908438 | 15.932862 | 84 | -0.069266 | |
| 7 | 0.3 | 0.05 | 17.869701 | 26.908438 | 15.706714 | 84 | -0.060465 | |
| 8 | 0.3 | 0.10 | 17.817334 | 26.908438 | 14.413428 | 78 | -0.059474 | |
| 9 | 0.4 | 0.01 | 17.526074 | 25.506203 | 12.074205 | 90 | -0.085047 | |
| 10 | 0.4 | 0.05 | 17.501982 | 25.506203 | 11.671378 | 65 | -0.087628 | |
| 11 | 0.4 | 0.10 | 17.501000 | 25.506203 | 10.929329 | 83 | -0.035040 | |

| | spearman_betweenness_vs_cascade | top10_overlap | top10_jaccard |
|---|---|---|---|
| 0 | 0.177899 | 0 | 0.000000 |
| 1 | 0.188079 | 0 | 0.000000 |
| 2 | 0.123189 | 0 | 0.000000 |
| 3 | -0.009017 | 1 | 0.052632 |
| 4 | -0.009886 | 1 | 0.052632 |
| 5 | -0.021118 | 1 | 0.052632 |
| 6 | -0.012214 | 0 | 0.000000 |
| 7 | -0.013817 | 0 | 0.000000 |
| 8 | 0.001938 | 0 | 0.000000 |
| 9 | -0.012003 | 0 | 0.000000 |
| 10 | -0.010017 | 0 | 0.000000 |
| 11 | 0.061694 | 0 | 0.000000 |


Mean Load Shed (%)
alpha
0.1
0.2
0.3
0.4
19.19
19.17
19.11
18.48
18.47
18.34
17.87
17.87
17.82
17.53
17.50
17.50
0.01
0.05
0.1
min_base_loading
mean_shed_percent
19.0
18.8
18.6
18.4
18.2
18.0
17.8
17.6


Spearman: Edge Betweenness vs Shed
alpha
0.1
0.2
0.3
0.4
0.08
0.09
0.05
-0.04
-0.05
-0.05
-0.07
-0.06
-0.06
-0.09
-0.09
-0.04
0.01
0.05
0.1
min_base_loading
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
spearman_betweenness_vs_shed




Top 10 edge-triggered cascades

| | initial_line | initial_lines | cascade_size | tripped_lines | iterations | solver_failed | lcc_size | served_load_mw | \ |
|---|---|---|---|---|---|---|---|---|---|
| 225 | 225 | [225] | 79 | [225, 179, 181, 220, 221, 0, 14, 16, 17, 18, 1... | 3 | False | 30 | 16283.52 | |
| 201 | 201 | [201] | 64 | [201, 145, 149, 150, 151, 153, 156, 157, 202, ... | 3 | False | 28 | 16847.52 | |
| 147 | 147 | [147] | 66 | [147, 145, 146, 149, 150, 151, 152, 153, 156, ... | 3 | False | 28 | 16987.82 | |
| 228 | 228 | [228] | 69 | [228, 254, 261, 262, 267, 268, 269, 29, 30, 31... | 4 | True | 34 | 17583.22 | |
| 269 | 269 | [269] | 74 | [269, 228, 257, 258, 259, 261, 263, 268, 29, 3... | 4 | True | 34 | 17586.22 | |
| 263 | 263 | [263] | 42 | [263, 228, 258, 260, 261, 265, 268, 269, 270, ... | 3 | True | 68 | 18007.22 | |
| 177 | 177 | [177] | 74 | [177, 57, 63, 159, 161, 176, 179, 196, 197, 19... | 3 | False | 25 | 18046.32 | |
| 99 | 99 | [99] | 81 | [99, 50, 91, 96, 102, 48, 49, 51, 65, 73, 90, ... | 7 | False | 34 | 18118.72 | |
| 262 | 262 | [262] | 46 | [262, 257, 258, 268, 270, 271, 0, 46, 52, 53, ... | 3 | False | 40 | 18144.42 | |
| 214 | 214 | [214] | 36 | [214, 213, 216, 217, 218, 17, 21, 52, 53, 55, ... | 3 | False | 82 | 18159.02 | |

| | shed_load_mw | shed_percent |
|---|---|---|
| 225 | 7564.13 | 31.718555 |
| 201 | 7000.13 | 29.353542 |
| 147 | 6859.83 | 28.765224 |
| 228 | 6264.43 | 26.268542 |
| 269 | 6261.43 | 26.255962 |
| 263 | 5840.43 | 24.490589 |
| 177 | 5801.33 | 24.326632 |
| 99 | 5728.93 | 24.023038 |
| 262 | 5703.23 | 23.915270 |
| 214 | 5688.63 | 23.854049 |

```
Top 10 node-triggered cascades
     initial_line    initial_lines  cascade_size                                   tripped_lines  iterations  solver_failed  lcc_size  served_load_mw  \
137           NaN       [201, 202]            67  [201, 202, 145, 146, 149, 150, 151, 153, 155, ...           3          False        28        16847.52
135         201.0            [201]            64  [201, 145, 149, 150, 151, 153, 156, 157, 202, ...           3          False        28        16847.52
116           NaN       [220, 225]            73  [220, 225, 0, 14, 16, 17, 18, 19, 20, 21, 22, ...           2          False        44        16899.82
95          147.0            [147]            66  [147, 145, 146, 149, 150, 151, 152, 153, 156, ...           3          False        28        16987.82
18            NaN     [27, 35, 36]            70  [27, 35, 36, 20, 22, 39, 41, 42, 16, 29, 30, 3...           4          False        34        17214.32
215           NaN       [259, 269]            64  [259, 269, 98, 228, 254, 258, 261, 262, 263, 2...           3          False        39        17335.22
114           NaN  [175, 176, 177]            85  [175, 176, 177, 159, 161, 170, 173, 174, 179, ...           3          False        25        17553.02
168         228.0            [228]            69  [228, 254, 261, 262, 267, 268, 269, 29, 30, 31...           4           True        34        17583.22
6             NaN     [15, 20, 21]            75  [15, 20, 21, 13, 16, 17, 19, 22, 24, 25, 27, 1...           4          False        37        17615.22
98            NaN  [146, 150, 152]            58  [146, 150, 152, 0, 17, 18, 20, 21, 22, 23, 24,...           2           True        59        17621.62

     shed_load_mw  shed_percent  initial_bus  incident_line_count
137       7000.13     29.353542          137                    2
135       7000.13     29.353542          135                    1
116       6947.83     29.134233          116                    2
95        6859.83     28.765224           95                    1
18        6633.33     27.815445           18                    3
215       6512.43     27.308477          215                    2
114       6294.63     26.395179          114                    3
168       6264.43     26.268542          168                    1
6         6232.43     26.134357            6                    3
98        6226.03     26.107520           98                    3
```

```
Node metric summary
             metric  pearson_vs_shed  spearman_vs_shed  pearson_vs_cascade  spearman_vs_cascade  top10_overlap_vs_shed  top10_jaccard_vs_shed  \
0            degree         0.405637          0.747169            0.545163             0.793540                      1               0.052632
2         closeness         0.226542          0.676267            0.395130             0.716611                      0               0.000000
3       eigenvector        -0.046565          0.658854            0.048297             0.700899                      0               0.000000
1  node_betweenness         0.123619          0.626020            0.282330             0.648111                      1               0.052632

   top10_overlap_vs_cascade  top10_jaccard_vs_cascade
0                         0                  0.000000
2                         0                  0.000000
3                         0                  0.000000
1                         1                  0.052632
```